

Algebraic Foundations

Simplifying, Expanding & Factorizing

The core manipulation skills for the whole course.

Use these notes well and you will be able to:

- Substitute values into expressions and formulas.
- Simplify by collecting like terms – single and multi-variable.
- Expand single brackets, pairs of brackets, and triple brackets (E).
- Factorise: HCF, grouping, difference of squares, perfect square, quadratic.
- Factorise cubic expressions of the form $ax^3 + bx^2 + cx$ (F 02).

If you follow the steps, you will succeed.



Introduction to Algebra

A letter (variable) stands for an unknown or general number. An **EXPRESSION** has no equals sign (e.g. $3x + 5$). An **EQUATION** has an equals sign (e.g. $3x + 5 = 14$). A **FORMULA** expresses a rule (e.g. $A = \pi r^2$).

TERM	EXAMPLE	WHAT IT MEANS
Expression	$4x^2 - 3y + 2$	No = sign. Cannot be 'solved', only simplified.
Equation	$4x - 3 = 9$	Has = sign. Can be solved for x.
Formula	$v = u + at$	A rule relating several variables.
Identity	$2(x+3) = 2x+6$	True for ALL values of x (symbol).

Substitute $x = 3$, $y = -2$ into $5x^2 - 4y + 1$

$$5(3^2) - 4(-2) + 1 = 5 \times 9 + 8 + 1 = 45 + 8 + 1 \\ = 54$$

Collecting Like Terms

LIKE TERMS have the same variable(s) and power(s). Only like terms can be added or subtracted.

	LIKE TERMS (can combine)	UNLIKE TERMS (cannot combine)
×	$3x + 5x = 8x$	$3x + 5x^2$ — different powers!
×	$7ab - 3ab = 4ab$	$7ab - 3a$ — different variables
×	$2a^2 - 6ab + 5a^2 + 3ab = 7a^2 - 3ab$	Order of variables doesn't matter: $ab=ba$

Simplify: $3x^2 + 2xy - x^2 - 7xy + 4x$

$$\text{Group like terms: } (3x^2 - x^2) + (2xy - 7xy) + 4x \\ = 2x^2 - 5xy + 4x \\ = 2x^2 - 5xy + 4x$$

Key Words

KEY WORD	SIMPLE DEFINITION
Variable	A letter representing an unknown or changing value.
Coefficient	The number multiplying a variable. In $7x^2$, the coefficient is 7.
Constant	A fixed number with no variable attached. e.g. the 5 in $3x + 5$.
Like terms	Terms with identical variable part. $3xy$ and $-7xy$ are like terms.
Expand	Multiply out brackets to remove them.
Factorise	Reverse of expanding — put back into bracket form.
HCF	Highest Common Factor — the largest expression dividing all terms.

Expanding Brackets

Single bracket — multiply each term inside by the factor outside

Expand $3x(2x - 4y)$

$$3x \times 2x = 6x^2$$

$$3x \times (-4y) = -12xy$$

$$= 6x^2 - 12xy$$

Two brackets (FOIL / grid) — each term in first $\times$ each term in second

Expand $(2x + 1)(x - 4)$

$$\text{F: } 2x \times x = 2x^2$$

$$\text{O: } 2x \times (-4) = -8x$$

$$\text{I: } 1 \times x = x$$

$$\text{L: } 1 \times (-4) = -4$$

$$\text{Collect: } 2x^2 - 8x + x - 4$$

$$= 2x^2 - 7x - 4$$

Tip: Special products to recognise: $(a + b)^2 = a^2 + 2ab + b^2$ | $(a - b)^2 = a^2 - 2ab + b^2$ | $(a+b)(a-b) = a^2 - b^2$
(difference of two squares)

Three brackets (F O2) — expand two first, then the third

Expand $(x - 2)(x + 3)(2x + 1)$

$$\text{Step 1: expand first two: } (x-2)(x+3) = x^2 + 3x - 2x - 6 = x^2 + x - 6$$

$$\text{Step 2: multiply result by } (2x+1):$$

$$(x^2+x-6)(2x+1) = 2x^3+x^2+2x^2+x-12x-6$$

$$\text{Collect: } 2x^3 + 3x^2 - 11x - 6$$

$$= 2x^3 + 3x^2 - 11x - 6$$

Factorising

Step 1 — Always try HCF first

Factorise fully: $9x^2 + 15xy$

$$\text{HCF of } 9x^2 \text{ and } 15xy = 3x$$

$$3x(3x + 5y)$$

$$= 3x(3x + 5y)$$

Factorising – All Methods

METHOD	FORM TO SPOT	EXAMPLE
HCF	All terms share factor	$12x^3 - 8x^2 = 4x^2(3x-2)$
Grouping	$ax+bx+kay+kby$	$xy+3x+2y+6 = x(y+3)+2(y+3) = (x+2)(y+3)$
Diff. of squares	a^2-b^2	$9x^2-16 = (3x+4)(3x-4)$
Perfect square	$a^2 \pm 2ab + b^2$	$x^2+6x+9 = (x+3)^2$
Quadratic ($a=1$)	x^2+bx+c	x^2+5x+6 : find p,q where $pq=6, p+q=5$ $(x+2)(x+3)$
Quadratic ($a \neq 1$)	ax^2+bx+c	$2x^2+5x+3$: split bx $2x^2+2x+3x+3$ $(2x+3)(x+1)$
Cubic (E)	ax^3+bx^2+cx	$x^3-3x^2+2x = x(x^2-3x+2) = x(x-1)(x-2)$

Factorise: $6x^2 - 7x - 3$

$ac = 6 \times (-3) = -18$. Find two numbers: product -18, sum -7 -9 and +2

Split: $6x^2 - 9x + 2x - 3$

Group: $3x(2x-3) + 1(2x-3)$

$= (3x+1)(2x-3)$

Factorise: $4a^2x^2 - 9b^2y^2$ (F 02)

Recognise: difference of two squares $a^2-b^2 = (a+b)(a-b)$

$= (2ax)^2 - (3by)^2$

$= (2ax + 3by)(2ax - 3by)$

$= (2ax + 3by)(2ax - 3by)$

Factorise: $3x^3 - 12x^2 + 9x$ (F 02)

HCF first: $3x(x^2 - 4x + 3)$

Factorise bracket: find p,q where $pq=3, p+q=-4$ -3 and -1

$3x(x-1)(x-3)$

$= 3x(x-1)(x-3)$

Practice Questions

SUBSTITUTION AND SIMPLIFYING

1. Evaluate $4p^2 - 3q$ when $p = -2$ and $q = 5$.

2. Simplify: $5m^2 + 3mn - 2m^2 - 7mn + m$

EXPANDING

3. Expand: $4y(3y^2 - 5y + 2)$

4. Expand and simplify: $(3x - 2)(2x + 5)$

5. Expand: $(x+1)(x-3)(x+4)$ [F 02]

FACTORISING

6. Factorise: $12a^2b - 8ab^2$

7. Factorise: $x^2 - 11x + 24$

8. Factorise: $3x^2 + 11x - 4$

ADD YOUR NOTES HERE



Hey smart student!

Factorising is the reverse of expanding — they are mirror images. Always take out the HCF first before trying any other method.

Recognise the 'difference of two squares' pattern instantly — it saves minutes in exams.

Keep going — see you in the next lesson! From FREDi